

# CODEGRAPH

## CodeGraph — Boba Code-Intelligence Integration

**Revision:** 1 **Last modified:** 2026-06-06T14:40:00Z **Authority:** Constitution §11.4.78 (CodeGraph mandate) · §11.4.79 (own-org submodules included, third-party excluded) · §11.4.80 (update/sync automation) · §11.4.10 (secrets never indexed) · §11.4.77 (regeneration mechanism)

CodeGraph (@colbymchenry/codegraph, <https://github.com/colbymchenry/codegraph>) is a 100%-local SQLite semantic knowledge graph of this codebase, exposed to AI coding agents over MCP. No cloud, no external API. It gives agents instant, consistent symbol / caller / callee / impact resolution instead of repeated shallow file scanning.

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### Install (§11.4.78)

```
npm install -g @colbymchenry/codegraph      # Node 18+, NO sudo
      (user-writable prefix)
codegraph --version                          # confirm the binary
      runs (exit 0 ≠ working binary)
```

On this host the binary resolves to /opt/homebrew/bin/codegraph (the npm-managed shim). A standalone-installer shim (~/.local/bin/codegraph) must NOT shadow it — if codegraph --version reports an older version than npm view @colbymchenry/codegraph version, a stale shim is winning on PATH; remove it so the npm-managed binary is active.

# Configuration model — zero-config (important)

CodeGraph 0.9.x is **zero-config**: there is **no** `.codegraph/config.json` **exclude list**. What gets indexed is decided by:

1. **Built-in defaults** — `dependency/build/cache` dirs (`node_modules`, `vendor`, `dist`, `build`, `target`, `.venv`, `Pods`, ...) are skipped even with no `.gitignore`, plus any file larger than **1 MB**.
2. **.gitignore** — honored directly (root + nested). **To keep something out of the index, add it to .gitignore**. To pull a default-excluded dir back in, add a negation (`!path/`).

(Source: the installed package `README.md` → “Configuration”).

## What is tracked vs ignored

| Path                                                              | Git        | CodeGraph index | Why                                                        |
|-------------------------------------------------------------------|------------|-----------------|------------------------------------------------------------|
| <code>.codegraph/codegraph.db (+ *.db-wal/*.db-shm/cache/)</code> | gitignored | n/a             | Build artifact; regenerate with codegraph index (§11.4.77) |
| <code>.mcp.json</code>                                            | tracked    | n/a             | MCP wiring, committed (§11.4.78)                           |
| <code>scripts/codegraph_validate.sh</code>                        | tracked    | indexed         | Anti-bluff validator (§11.4.78 step 4)                     |
| <code>docs/codegraph/Status.md</code>                             | tracked    | n/a             | Append-only event ledger (§11.4.80)                        |

## Exclusions — secrets (§11.4.10) + third-party (§11.4.79)

**Secrets are never indexed.** Root .gitignore already excludes .env, .env.\*, \*.env, .qbit.env, \*secrets\*, config/qBittorrent/, config/jackett/, config/boba.db\*. Because CodeGraph honors .gitignore, none of these reach the index. Audited: 0 real secret paths in the index.

**Third-party submodule excluded.** submodules/jackett (upstream Jackett/Jackett, C#/.NET) is third-party — §11.4.79 requires it OUT of the index. CodeGraph 0.9.9 does index C#, so an explicit .gitignore entry submodules/jackett/ was added. This is **inert for git** (the submodule is a tracked gitlink via .gitmodules; an ignore entry on a tracked path does not un-track it) but tells CodeGraph to skip it. Audited: 0 submodules/jackett paths in the index (down from 890 before the exclusion).

**Own-org submodule INCLUDED.** constitution/ (HelixDevelopment, own-org per §11.4.79) is deliberately NOT ignored, so its Go workable-items sources are indexed and resolve cross-submodule (e.g. versionTagsCmd).

## Index / re-generate

```
codegraph index .           # full (re)index – the §11.4.77
                             regeneration mechanism
codegraph index . --force   # force a clean re-index
codegraph sync .           # incremental update since last index
codegraph status .         # files / nodes / edges
```

Current index: **509 files, 8906 nodes, 17025 edges** (go, python, typescript, properties, xml, yaml).

## MCP wiring (Claude Code, project-scoped + committed)

.mcp.json at the repo root registers the CodeGraph MCP server over stdio:

```
{
  "mcpServers": {
    "codegraph": { "command": "codegraph", "args": ["serve", "--mcp"], "env": {} }
  }
}
```

```
}  
}
```

serve --mcp is the stdio transport; the command is the bare codegraph on PATH (no hardcoded host path, per §11.4.78). It exposes 8 tools: codegraph\_search, codegraph\_callers, codegraph\_callees, codegraph\_impact, codegraph\_node, codegraph\_explore, codegraph\_status, codegraph\_files.

Other agents: register the same codegraph serve --mcp stdio server via that agent's MCP config (Cursor, opencode opencode.json, Qwen .qwen/settings.json, Codex, etc.) — codegraph install automates this for supported agents.

## Anti-bluff verification (§11.4.78 step 4)

```
bash scripts/codegraph_validate.sh          # 7 checks; exit 0 =  
all PASS
```

The validator proves the index is real and correctly scoped — six checks plus an **unforgeable MCP challenge**: it drives the real codegraph serve --mcp server and asserts the codegraph\_status node count returned over MCP equals the CLI node count. An agent answering from its own file-reading tools cannot fabricate that number. The other checks assert: index reality (nodes > 0), own-code symbol resolution, own-org constitution/ symbol resolution (§11.4.79 included), and zero submodules/jackett / zero secret paths in the index (§11.4.79 + §11.4.10 excluded). The path-audit detection logic is falsifiability-rehearsed (a synthetic leaked path is detected) so the green PASS is not a tautology.

## Update + sync automation + cadence (§11.4.80)

Scripts are **inherited by reference** from the constitution submodule — never copied:

```
bash constitution/scripts/codegraph_update.sh  
# npm-update to latest + $107 version check  
bash constitution/scripts/codegraph_sync.sh .  
# status → sync → status → validate, logged
```

`codegraph_sync.sh` appends every run to BOTH this project's docs/  
`codegraph/Status.md` and the constitution's own ledger. **Cadence:**  
weekly floor (\$11.4.45 status-digest cadence); may run more often.

## Troubleshooting

- **codegraph reports an old version after npm update** — a stale standalone shim on PATH (commonly `~/.local/bin/codegraph`) is shadowing the npm-managed binary. Remove the stale shim.
- **A path you expect is missing from the index** — check it isn't `.gitignored`, default-excluded (`node_modules/dist/...`), > 1 MB, or an unsupported language.
- **Third-party code appears in the index** — add its path to `.gitignore` (per the zero-config exclude model) and `codegraph index . --force`.